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Comparative Summary and Analysis

Comparative Summary (Short)

The five selected research papers demonstrate how machine learning and deep learning techniques are improving electricity consumption prediction, bill analysis, tariff recommendation, and energy management. Lazzeroni et al. (2023) reconstructed hourly load profiles using k-NN regression, while Han et al. (2024) achieved 98% accuracy in electricity bill recovery risk prediction using a Bi-LSTM model. Wang et al. (2024) developed a hybrid Self-Attention, LSTM, RNN, and XGBoost model for accurate urban electricity demand forecasting. Jeong and Joo (2024) recommended personalized electricity tariffs using deep learning matrix factorization, even without historical smart-meter data. Nigar et al. (2024) proposed an Energy Advisor system that used Random Forest to predict electricity consumption with an R^2 score of 0.98 while providing personalized energy-saving recommendations.

Overall, these studies show that AI-based models provide higher prediction accuracy and better energy management than traditional methods. However, challenges such as limited datasets, scalability, and real-world deployment still exist. These research gaps motivate the development of **WattWise**, which aims to deliver accurate electricity bill prediction along with personalized energy-saving recommendations using machine learning.

Comparative Analysis of Research Papers

Paper	Problem	Data Input	Method	Main Finding	Key Limitation / Caveat
Lazzeroni et al., 2023	Predict hourly load profiles for unobserved users	Monthly ToU bills and limited hourly measurements	k-NN regression framed as an inverse problem	~26% instantaneous error and <4% ToU error; outperformed standard methods.	Future work: richer metrics, seasonal models, advanced ML.
Han et al., 2024	Predict electricity bill recovery risk	Historical billing and smart-meter data	Bidirectional LSTM	~98% classification accuracy.	Needs larger and more comprehensive datasets.
Wang et al., 2024	Predict monthly urban electricity consumption	Electricity, weather, and social data	Self-Attention + LSTM + RNN + XGBoost	Improved long-term forecasting accuracy and generalization.	Future work: small-sample learning and efficient deployment.
Jeong & Joo, 2024	Recommended personalized tariffs	Customer profiles, estimated usage, past bills	Deep Learning Matrix Factorization	Recommended TOU tariffs with lower prediction error.	Depends on quality of estimated consumption patterns.
Nigar et al., 2024	Demand-side energy management and consumption prediction	Historical consumption and environmental factors	ML Energy Advisor (Random Forest, Decision Tree, etc.)	Random Forest achieved $R^2 \approx 0.98$ and personalized energy-saving recommendations.	Needs validation on broader datasets and real-world deployment.

REFERENCES:

PAPER-1: A Monthly Urban Electricity Consumption Prediction Algorithm Based on Self-Attention

LINK: <https://ieeexplore.ieee.org/document/10485739>

PAPER-2: Artificial Intelligence Based Risk Prediction System for Electricity Bill Recovery

LINK: <https://ieeexplore.ieee.org/document/10548630>

PAPER-3: .A Data-Driven Approach to Predict Hourly Load Profiles From Time-of-Use Electricity Bills

LINK: <https://onlinelibrary.wiley.com/doi/full/10.1155/2024/6339681>

PAPER-4: .Personalized Electricity Tariff Recommendation Method for Residential Customers Lacking Historical Metering Data Incorporating Customer Profiles and Behavioral Changes

LINK: <https://ieeexplore.ieee.org/document/10520307>

PAPER-5: Improving Demand-Side Energy Management With Energy Advisor using machine learning

LINK: <https://ieeexplore.ieee.org/document/10151868>